

RESEARCH CONTRIBUTION: E-COMMERCE BEHAVIORAL DYNAMICS

Milestone 6 - Advanced Statistical Interpretation

Abstract: This research paper provides an advanced synthesis of customer behavior within a digital marketplace. By applying normalization techniques and linear regression modeling, we evaluate the predictive power of various features. Findings suggest that engagement metrics are more indicative of fiscal contribution than demographic markers alone.

1. Methodology and Data Preparation

The study utilized an ETL pipeline to standardize features using Z-score normalization. This ensures that features with different scales—such as Age and Total Spend—are weighted equally within the predictive model. The normalization formula used is:

$$Z = (x - \mu) / \sigma$$

2. Advanced Statistical Findings

Data analysis indicates that "Items Purchased" is the primary driver of revenue. However, a deeper look at membership tiers reveals that Gold members maintain high spending levels regardless of minor fluctuations in satisfaction, whereas Bronze members are significantly more "satisfaction-elastic," showing a sharp decline in spend following a poor rating.

3. Conclusion and Recommendations

The project concludes that interactive Decision Support Systems significantly reduce the latency between data acquisition and strategic action. Future work should focus on integrating external economic indicators to improve model accuracy across lower-tier customer segments.